From Sky to Sea: The Biomimetic Evolution of Chiroptera

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Abstract. The Chiroptera leisure sailboat merges biological principles with modern sailcraft engineering to produce a vessel that performs efficiently, deploys easily, offers enhanced spatial comfort, and stores compactly. The design draws from the folding mechanics of bat wings, the rapid movement of nastic plant structures, and the hydrodynamic stability of whales, with supporting insights from flying fish and duck feet. The resulting system integrates a deformable adaptive sail with a stable, spacious hull. This report documents the design process, evaluates the quality of the biological analogies, compares the concept against existing competitors, and reports prototype findings and future directions. A three-segment folding mast deployed by a single pull wire reduced stowed height substantially relative to a fixed spar, and scaled water testing confirmed that the bat-inspired sail achieved flotation and movement comparable to a conventional rig.

Keywords. biomimicry; bio-inspired design; deployable structures; sail systems; hull hydrodynamics; bat wing morphology; folding mechanisms

1 Organism selection

The team began by examining organisms known for efficient movement through air or water. Bats became the central inspiration because of the combination of skeletal flexibility, membrane adaptability and aerodynamic control in their wings.

Figure 1. Underside of a bat wing showing skeletal–membrane structure and joint articulation, which served as the morphological reference for the adaptive sail rib design in the Chiroptera mast.

Bat wings use elongated finger bones that alter shape in real time, producing fine-tuned camber and lift adjustments throughout the wingbeat cycle. This capability has been studied for its potential application to engineered flight and deformable structures [6]. The structural strategy translated directly into the sail, guiding the design of ribbed membranes, jointed masts and adaptive curvature. Three properties of bat flight became design goals in their own right: stability under variable forces, compact storage through folding, and dynamic surface shaping.

Complementary sources broadened the search. Flying fish demonstrate how flexible gliding fins extend movement efficiency across water and air, illustrating the effect of membrane properties on lift and drag reduction [5]. Duck feet provide an example of modulated surface area optimising underwater thrust and manoeuvring [7]. Rapidly folding plant structures, particularly *Mimosa pudica*, offered a model for deployable mechanisms, suggesting simple repeatable motions applicable to reliable sail deployment [4]. The whale contributed a hull model, demonstrating how volume and curvature can be balanced to provide both stability and spaciousness; its broad midsection and tapered ends describe a geometry that optimises flow while accommodating substantial internal capacity. Together these established a framework for translating aerial principles into a marine context.

2 Concept selection

Narrowing the focus to bats and whales relied on structured evaluation across four abilities: lifting, adaptability, folding and space efficiency. Among the initial sketch concepts, designs using bat wings as inspiration proved most promising. Functional matrices, Shah metrics, Helms metrics, a T-chart and a Pugh matrix were used to assess the designs quantitatively, and the concepts scoring highly across all categories were carried forward.

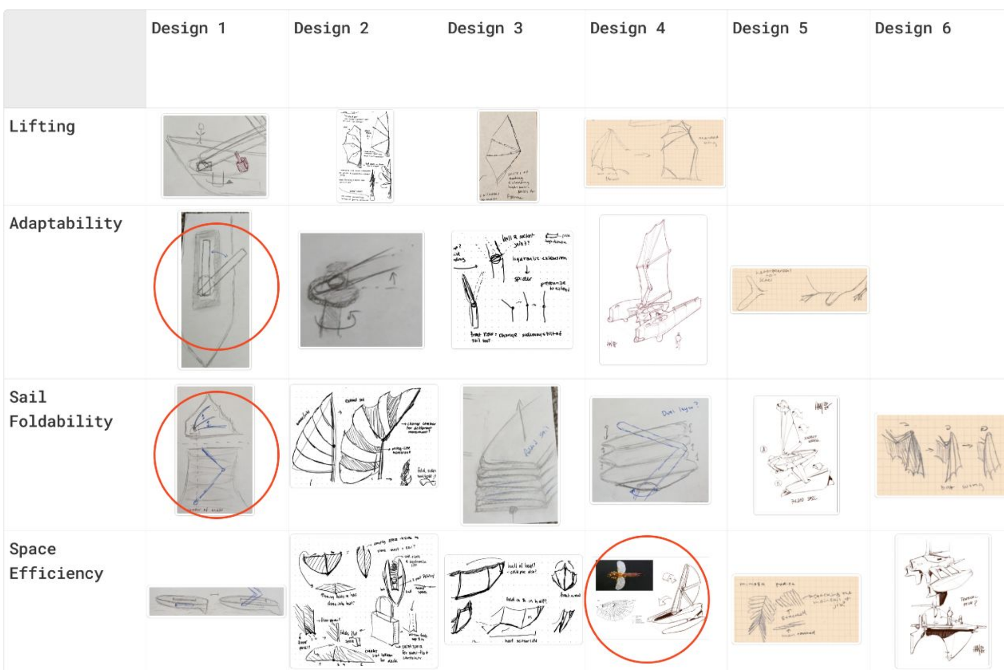


Figure 2. Functional matrix containing the initial design sketches and the concepts selected to move forward, scored across lifting, adaptability, sail foldability and space efficiency.

With bat wings providing structure for the sail and mast, whales supplied a complementary framework for the hull, demonstrating that elegance in form can coexist with practical performance. This systematic approach identified strategies that were both ambitious and achievable.

3 Competitive landscape

Existing systems established which user needs remained unfulfilled. Balogh Sail Designs' BATWING sail incorporates bat-inspired battens but does not translate the adaptive shape-changing properties of biological wings [2]. Michelin's inflatable sails and modular rigs such as the junk rig demonstrate useful attempts at deformation control and reduced manual trimming, but lack the capacity for multi-jointed adaptation found in natural wings. The Transition Rig, inspired by avian wings, introduces biomorphic joints but was created primarily for smaller craft such as kayaks and windsurfers [8], and flexes only partially in response to heavy wind rather than collapsing and folding completely.

(a) (b)

Figure 3. Two existing biologically inspired sail systems. (a) The BATWING sail, which employs bat-inspired battens to support membrane curvature. (b) The Transition Rig, which uses an avian-inspired articulated geometry. These references illustrate current approaches to nature-inspired rigging and establish the design landscape for Chiroptera. Sources: BSD [2]; Transition Rig [8].

Collectively these systems reveal growing interest in biomimicry while also highlighting the absence of a sail system that meaningfully reproduces the dynamic control and space efficiency of natural wings. That gap set the design priorities.

4 Final concept

The sail uses bat-like fingers constructed from PLA and tensioned ripstop nylon. These ribs allow the membrane to deform under wind load, creating a continuously adaptive airfoil rather than a fixed cut. The mast folds in three segments mirroring the primary joint structure of a bat wing, and is deployed through a single pull wire mechanism that extends or retracts the structure through a pulley system.

The membrane is supported by a continuous lattice of flexible elements that distributes stress evenly, preserves airfoil shape and reduces flutter. The hull, inspired by whales, uses gradual curvature to promote laminar flow and maintain stability during manoeuvres.

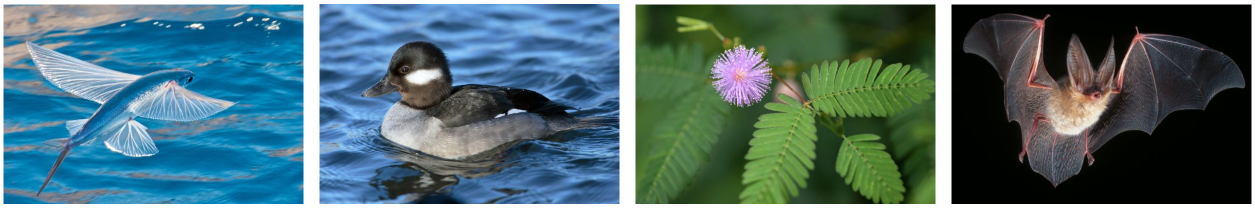


Figure 4. Mast deployment sequence, left to right: folded into a compact boom, mid-deployment, and fully extended to demonstrate the bat wing structure. The mechanism is driven by a single string pull and a tension-based unfolding action.

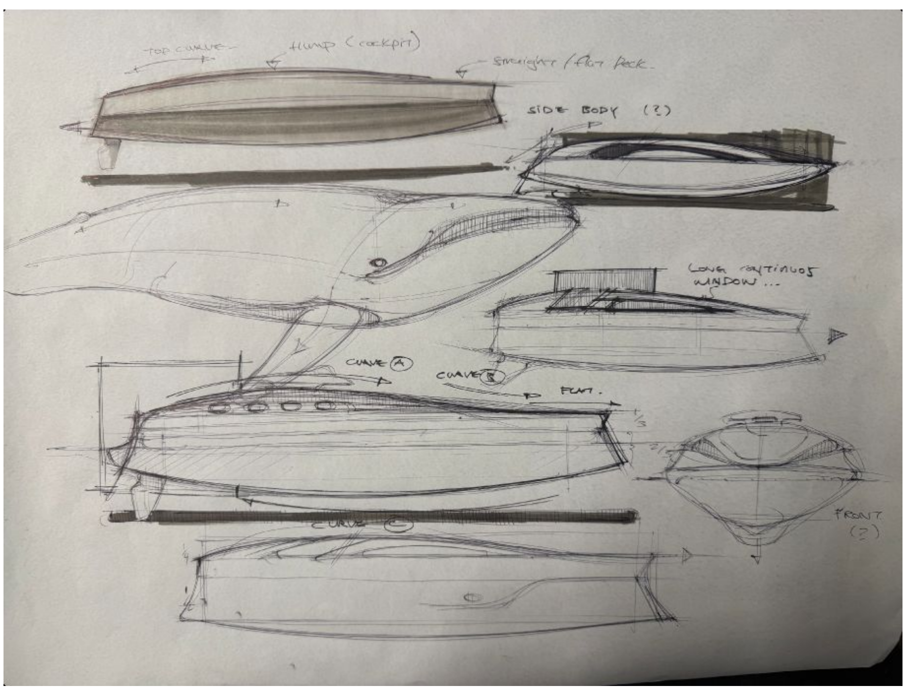
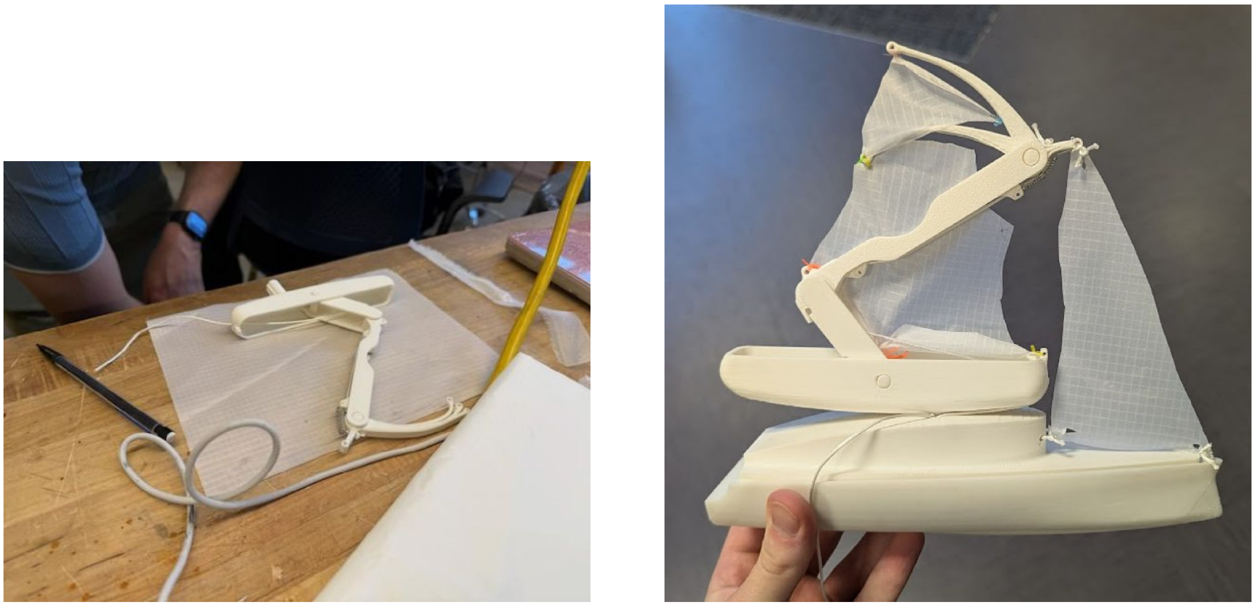


Figure 5. Hull development. The side profile echoes the fluid silhouette of a whale, prioritising spaciousness and stability while limiting drag.

5 Prototype and testing

Prototype fabrication used 3D printed PLA for the mast and boom and resin printing for the hull, reducing water permeability while remaining lightweight. The sail membrane was produced from ripstop nylon, which offered appropriate strength and flexibility at model scale.



(a) (b)

Figure 6. Physical prototype in PLA and ripstop nylon. (a) Mast extended. (b) Floating during water testing.

Pool tests at the Campus Recreation Center used a fan to simulate wind, allowing the adaptive sail's performance to be observed directly. These early tests showed that the bat-inspired sail achieved flotation and movement similar to a traditional sailboat, though further testing is required to quantify the comparison.

6 Refinements required

Although the prototype demonstrated feasibility, several improvements are needed before the next design phase.

- **Joint durability.** The mast has many joints and will experience stress concentration at the interfaces, which suggests more resilient materials such as carbon fibre composites or reinforced PLA blends.
- **Buoyancy.** The hull requires further optimisation through both geometry and ballast distribution, particularly if the full-scale vessel is expected to support significant interior volume.
- **Flow behaviour.** Additional CFD analysis will refine hull curvature to reduce upstream turbulence and improve wake behaviour for better vortex generation.
- **Membrane tension.** The sail requires better tensioning, possibly through adjustable anchors, elastic reinforcement, or different materials once scaled up.

These refinements would establish a more durable and testable intermediate prototype.

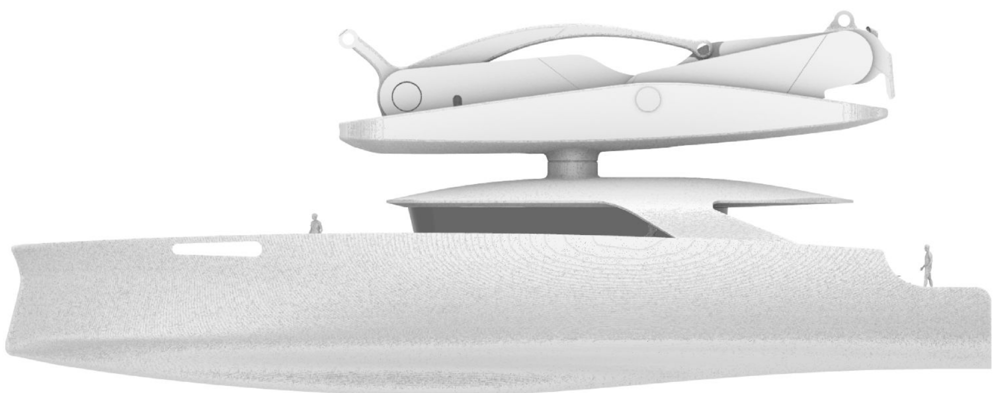


Figure 7. Side profile of the full-scale design with the mast collapsed, showing the refined viewing area and human figures for scale. Collapsing the rig into the boom is what reduces stowed height and expands compatibility with low-clearance berths.

(a) **(b)**

Figure 8. In-scene renderings of Chiroptera with the rig deployed, demonstrating the vessel's form and visual presence in realistic sailing environments.

7 Future work and commercial framing

Future testing will compare the Chiroptera system directly against a traditional sailboat under matched wind and hydrodynamic conditions, producing objective performance data that can validate the biological inspirations used in the design.

Beyond performance, the concept has a clear commercial argument. A folding mast dramatically reduces vessel height, which lowers storage costs and expands compatibility with areas of limited clearance. The whale-inspired hull offers spacious interiors that appeal to leisure markets. The organic forms derived from bat and whale morphologies also create a distinctive aesthetic profile suited to both recreational and exhibition contexts.

8 Conclusion

The design process for Chiroptera demonstrates the potential of biomimetic design to turn ambitious concepts into tangible engineering solutions. Translating the adaptability, precision and efficiency observed in flight into a maritime context produced a vessel that combines performance, deployability and spatial elegance.

Biomimicry demands more than surface-level replication. It requires abstracting biological principles into actionable engineering strategies. The bat wing integrates flexibility, structural support and control, qualities that cannot be copied literally but can inspire functional equivalents in mechanical design. The most successful biomimetic outcomes arise when biological insight is combined with careful abstraction, iterative prototyping and rigorous testing. The interdisciplinary range of this project, spanning biology, mechanical engineering, materials science and industrial design, strengthened both the technical depth and the creative reach of the final concept. Continued work will refine the structural mechanisms, improve the reliability of the folding system, and conduct higher-fidelity aerodynamic and hydrodynamic testing.

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Document note. Figures 2, 4, 5, 6, 7 and 8 are original work. Figure 1 and Figure 3 reproduce third-party photographs credited in the reference list. Pool testing was conducted at model scale under fan-generated wind; no full-scale vessel was built, and the performance comparison against a conventional rig remains to be carried out.